Spatial Transcriptomics Inference for the Elucidation of Disease Pathogenesis Across Large Scale Histopathology Cohorts: A Preliminary Analysis in Skin Photoaging

Gokul Srinivasan[1], Matt Davis MD[2] , Matt LeBoeuf MD PhD[2], Josh Levy PhD[1]
[1]Cedars Sinai Medical Center, Los Angeles, CA, [2]Dartmouth Health, Lebanon, NH

Cedars Sinai

ABSTRACT

- **Spatial transcriptomics** (ST) methods enable **insight** into disease pathogenesis across histopathological slides.
- However, ST assays are **resource intensive** both financially and temporally, and require highly skilled technicians.
- **ST inference** methods are poised to **bridge existing gaps** in access by directly inferring ST data from widely available H&E slides.
- We provide **preliminary evidence** that ST inference methods can **accurately recapitulate natural genetic heterogeneity** and may be used to progress disease understanding via large scale histopathological cohorts.

INTRODUCTION

- **Spatial Transcriptomics:**
  - ST methods have been used to deepen insight into disease etiopathogenesis across a wide range of indications including diabetic foot ulcers, oral squamous cell carcinoma, and metastatic brain tumours.
  - Existing assays vary with respect to spatial resolution and marker multiplexing. Some assays (e.g.,Visium 10X Genomics assay) providing high multiplexing and low spatial resolution. Others (e.g., MERFISH) provide higher spatial resolution with relatively lower spatial multiplexing.
  - ST methods are uniquely positioned to accelerate drug effect discovery, help uncover novel therapeutic targets, and usher in age of personalized medicine (**Figure 1B**).
- **Challenges to Broader Adoption:**
  - ST adoption is currently limited on three fronts: 1) cost, 2) access to skilled technicians, and 3) challenging analytics.
  - Initial studies have demonstrated ability to infer ST from H&E Whole Slide Images (WSI), but have not been validated at scale

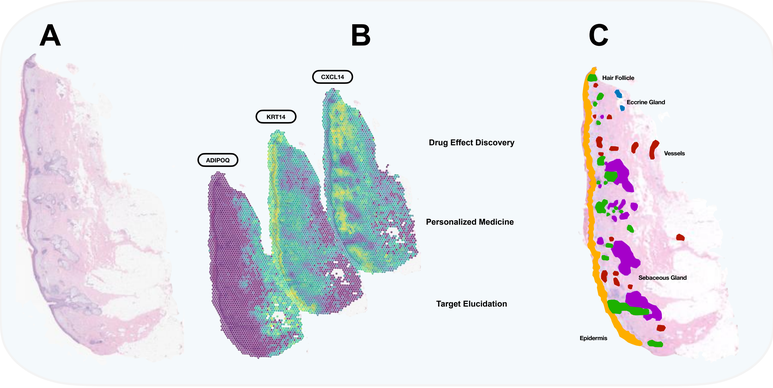

Figure 1: *Spatial Transcriptomics Overview.* (A) H&E skin histopathology tissue section. (B) Visium marker genes and ST utility. (C) Tissue image colored by histology type.

METHODS

- **Goal:** Establish feasibility and benchmark the precision of ST inference methods across large scale histopathological cohorts.
- **Data Acquisition and Preprocessing**
  - 261 tissue samples obtained in a single site Mohs micrographic surgery clinic. The samples were mostly from the head and neck, and they were graded for solar elastosis (SE) severity.
  - Formalin fixed specimens were processed, sectioned and stained for hematoxylin and eosin (H&E) using automated tissue staining instruments for subsequent imaging at 40x resolution (0.25 micron/pixel) using Aperio GT450 scanners.
  - ST ground truth data was obtained using four specimens from this larger cohort (two mild SE, two severe SE) via the Visium ST assay.
- **Model Development and Validation**
  - A SWIN transformer was trained to infer the log-adjusted count-normalized gene expression from 512 by 512-pixel histopathology image patches. The gene expression profile for a given image patch was taken to be equivalent to the gene expression profile of the Visium spot at its center (**Figure 2A**).
  - Models were trained to predict the top 1000 spatially variable genes concurrently, where these genes were determined using the SpatialDE package.
  - Models achieved a macro-averaged median Spearman's R value of 0.600 (CI: 0.580-0.610) across four-fold cross-validation.
  - Extensive analysis revealed that models recapitulated GT genetic heterogeneity both histologically and via pathway analysis.
- **Inference Workflow**
  - Trained models were applied across novel tissue sections as follows: 1) tissue detection and segmentation via SAM model, 2) patch generation and normalization, 3) inference (**Figure 2B**).

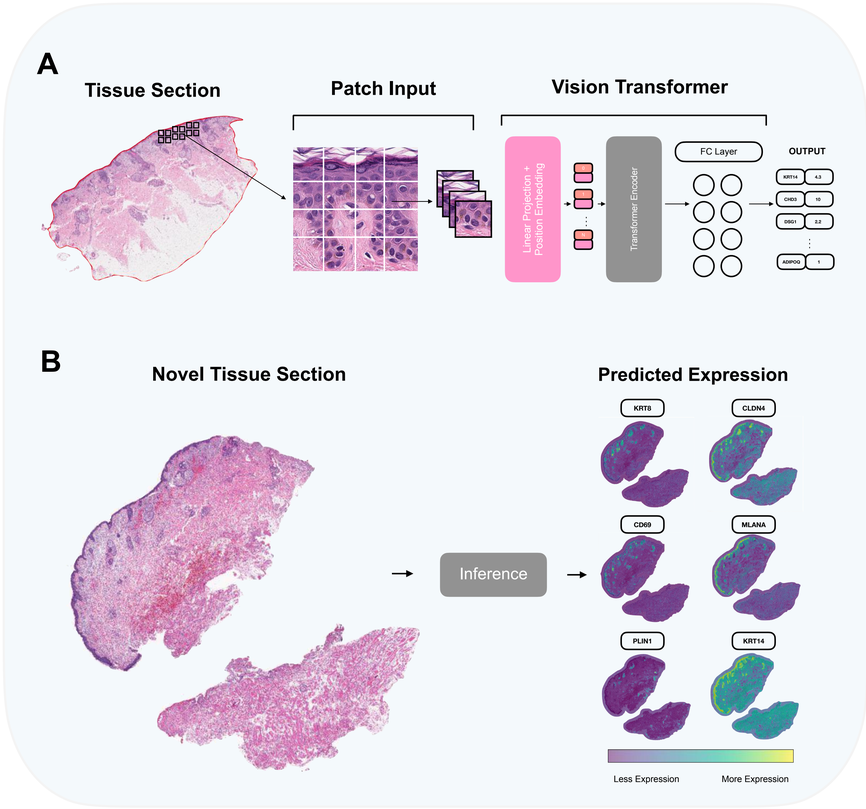

Figure 2: *Spatial Transcriptomics Inference Workflow.* (A) Inference task design. (B) Inference on novel tissue section.

RESULTS

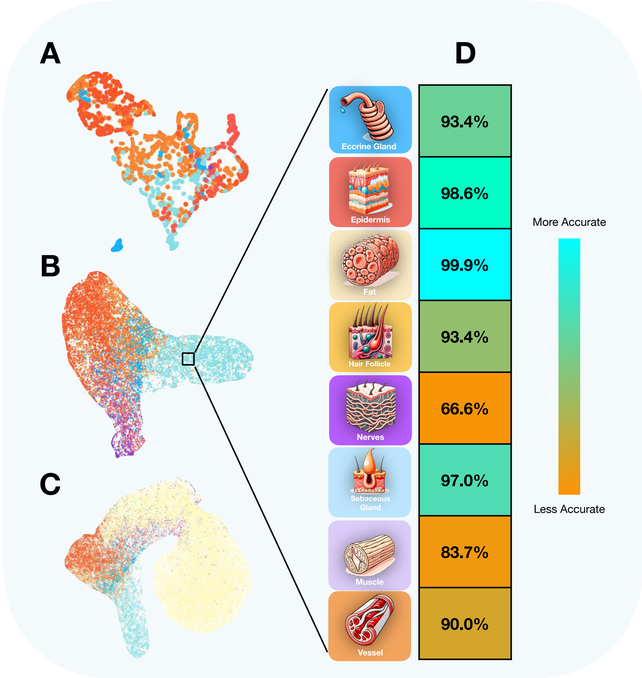

Figure 3: *Inference Methods Recapitulate Genetic Heterogeneity.* (A) Ground truth ST UMAP colored by histology type. (B) Inferred ST UMAP colored by histology type (-fat). (C) Inferred ST UMAP colored by histology type (+fat). (D) Histology prediction accuracy using inferred data.

CONCLUSION

- **Potential Value of ST Inference:**
  - ST inference methods enable the prediction of histological identity from spot-level gene expression, providing preliminary evidence that such inference methods can be used in the future to study other biological phenomena (e.g., skin photocarcinogenesis) at scale (**Figure 3**).
- **Limitations:**
  - Inference models trained using skin histopathology data leveraged a relatively homogenous training cohort (4 women, 2 mild + 2 severe SE), limiting sex-specific histomorphological generalization.
  - Models were trained to predict the top 1000 spatially variable genes, thus failing to account for several canonical markers that have less distinct spatial patterning.
- **Future Directions:**
  - Validate results using IHC staining + ancillary proteomics and fluorescence techniques and train models using more data and across increasingly varied gene target distributions.
- **Data and Code Availability:**
  - Data available on reasonable request, privacy/ethical restrictions.
- **Funding:**
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